PROJECT DOCUMENT

# BIOMASS ENERGY & AI

*Reforesting in Mitsue*

*Business Case & Decision Framework*


| | |
|---|---|
| **Version** | v1.6 |
| **Date** | 2026-07-17 |
| **Author** | Rob Oudendijk |

# BIOMASS ENERGY & AI — Business Case & Decision Framework

## Reforesting in Mitsue

## The "Why Say Yes" Layer

> **"The forest our ancestors planted — the power that sustains the village they built"** 先人が植えた森が、今、村を支える力になる。

*This document sits **above** the Implementation Plan. The implementation plan answers "what we will build and when." This document answers the question every decision-maker actually asks: **"Why should I say yes, and what do I get?"** It consolidates the financial and decision logic scattered across the revenue model, funding flowchart, EVM, and risk register, and fills the three gaps those documents leave open: (1) the entity & financing structure, (2) the alternatives the village is really choosing between, and (3) whole-project return, residual risk, and decision authority.*

*All financial figures are illustrative early-stage estimates consistent with `mitsue_revenue_model.md` and `mitsue_evm_plan.md`. Phase 1 feasibility studies (due M9, Dec 2026) replace them with vetted figures.*

# 1. Entity & Financing Structure — what each funder gets

The project raises money three ways — **grants, equity, and revenue/energy-share (no debt)**. These cannot all live in one organisation: a Japanese non-profit (一般社団法人 / NPO法人) **cannot issue equity or distribute profit to investors**. The project therefore uses a deliberate **two-tier structure**, which is also its mission-protection mechanism.

## 1.1 The two tiers

| | Tier 1 — Mission entity | Tier 2 — Operating company |
|---|---|---|
| Form | 一般社団法人 (non-profit) | 合同会社 GK (for-profit) |
| Holds | Charter, 官民連携 relationship with village, brand, IP/"playbook", forestry-restoration programme, J-Credit registration | Data-center assets, energy plant (biomass CHP, solar, battery), grid/FIT contracts, hosting contracts, staff |
| Receives | Founder capital, government grants, foundation grants — **all non-distributing** | Equity investment, revenue/energy-share contracts, operating revenue |
| Controls | **Owns the majority of the GK** + holds a charter "mission-lock" clause | Operated under contract to the mission entity |
| Returns to funders | None — return *is* the mission (public benefit) | Dividends / revenue-share from operating surplus |

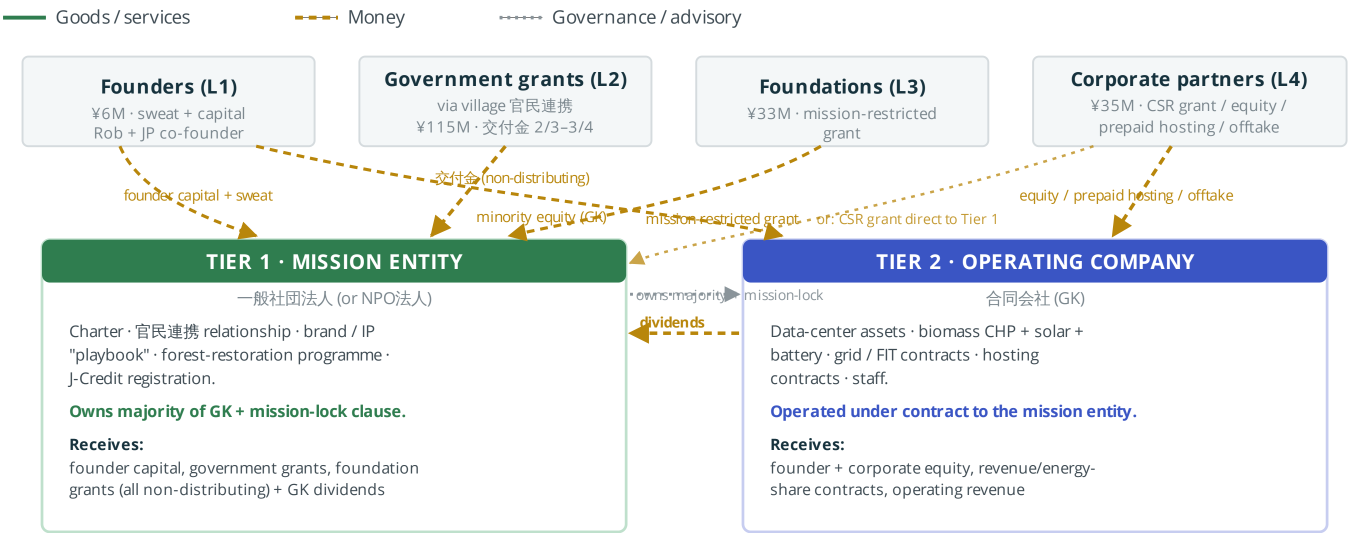
The mission entity owns and controls the operating company. Grant money (which legally cannot generate private profit) stays in Tier 1 or funds shared infrastructure; investor money that expects a return sits in Tier 2, where returns are legal and contractual. The 交付金 (MoE decarbonization grant) flows to the **village**, which co-invests the subsidized capex into the shared energy assets via the 官民連携 partnership.

## 1.2 What each funding layer is, and what it gets back

| Layer | Source | ¥M | Instrument | What they get |
|---|---|---|---|---|
| L1 | Founders (Rob + JP co-founder) | 6 | Founder equity in GK + sweat | Minority equity; mission stewardship |
| L2 | Government grants (national/pref./village) incl. **交付金 2/3–3/4 capex subsidy** | 115 | Non-refundable grant (to village / Tier 1) | RE-plan targets met; depopulation/carbon outcomes; no financial return expected |
| L3 | Foundations | 33 | Mission-restricted grant | SDG/impact outcomes; reporting; brand association |
| L4 | Corporate partners | 35 | **Choice of:** CSR grant · minority GK equity · prepaid hosting (revenue advance) · energy-share offtake | Green-compute capacity, ESG story, or contracted return depending on instrument chosen |
| L5 | Operating revenue | 3+ | Reinvested surplus | — |
| **Gap** | To close ¥28–53M | 28–53 | Additional 交付金 + corporate GK equity + revenue-financing | As per L2/L4 above |

## 1.2a How money and goods actually move between the two tiers

(Added 2026-07-17.) The tables above describe *what each layer gets*; this shows *how it flows* — goods/services (green, solid) vs. money (gold, dashed) vs. governance (grey, dotted) — between the two real legal entities.




*Reading it: every money line into Tier 1 is non-distributing by law — it funds mission activity, not investor return. Every money line into Tier 2 is a real financial instrument (equity, revenue-share) because Tier 2 is where legal profit distribution is possible at all. The only money flowing back **out** of Tier 2 is to Tier 1 (dividends, since Tier 1 owns the majority) — there is no path for grant money to leak into a private return, and no path for investor capital to bypass Tier 2's contractual bounds.*

## 1.3 Mission-lock — why investors cannot capture the project

Because equity exists only in Tier 2 and the Tier-1 non-profit holds the majority plus a charter clause, investors get a **real but minority** return and **cannot redirect the mission, sell the forest programme, or strip the community assets**. This is the answer to "what am I committing to / what do you promise": grant funders commit to a public good and are promised outcomes; equity/revenue-share investors commit capital to Tier 2 and are promised a contracted share of a defined operating surplus — bounded so the village always retains control.

### 1.4 Return profile (illustrative, base case)

Operating surplus accrues in the GK: **net surplus ¥10–32M/yr by Year 5**, project-level capital payback **10–18 years**. Equity-investor return depends on entry valuation and share, both set in Phase 2 once feasibility figures exist — **this document defines the *structure*; Phase 2 sets the *price*.** No instrument promises a return the operating surplus cannot support.

## 2. Alternatives Analysis — what the village is really choosing between

Every decision-maker has a "do nothing" option. For the village it is concrete: **underused space stays underused.** The project will choose its premises from **several candidate sites**, all underused village building stock. One strong candidate is the **former Sugano Elementary School (Koryukan / みつえ体験交流館)** — a former wooden elementary school that today operates under the name Mitsue Taiken Koryukan as a cultural and educational facility but has empty, underused classrooms; here the hub would occupy spare capacity and **complement** the Koryukan's existing crafts-and-experience role rather than replace it. Disused former factory or workshop buildings are other candidates. Final site selection is a Phase 1 decision.

> **Note on the dome school:** the dome school is a *separate* building, already in its own bidding/selection process to be reused by another party — it is **not** part of this project and should not be confused with the Koryukan.

A proposal is only persuasive against the alternatives it beats. For the candidate premises the village is realistically choosing among five paths:

| Option | Upfront cost to village | Ongoing | Jobs | Energy resilience | Forest / ecology | Productive reuse | Unlocks 交付金 | Reversible |
|---|---|---|---|---|---|---|---|---|
| **A. Status quo — spare capacity idle** | ¥0 now | Ongoing upkeep, no new return | 0 | None | Forest untended (liability grows) | No new use | No | Yes |
| **B. Solar-only** | Low–med capex | FIT income, thin | ~0 | Intermittent only | None | Partial | Partly (no resilient site) | Yes |
| **C. Forestry-only** | Low | J-Credit + timber, slow | Few | None | Restored ✓ | No | No | Yes |
| **D. Sell / lease to another party** | One-time cash or rent | Control lost | Uncertain | None | Lost | Buyer's choice[1] | No | Largely no |

| Option | Upfront cost to village | Ongoing | Jobs | Energy resilience | Forest / ecology | Productive reuse | Unlocks 交付金 | Reversible |
|---|---|---|---|---|---|---|---|---|
| **E. Integrated project** (proposed) | Mostly grant-funded² | Net surplus by ~Y5 | 8–15 | 24/7 (biomass CHP island-mode) | Restored ✓ | Yes ✓ | **Yes ✓** | Staged / descopable |

¹ Where a private buyer is involved, the village should weigh buyer credibility — the dome-school bidding file shows prospective-buyer quality varies; selling or leasing does not guarantee a good village outcome. ² Village cash exposure is minimized because the 交付金 subsidizes 2/3–3/4 of eligible energy capex and other layers carry the rest.

## Why the integrated project wins

- **It is the only option that satisfies the village's *own* published RE plan.** That plan's "one resilient renewable + storage + EV site" indicator currently reads zero; only Option E provides the "1". Options A–D leave the village's adopted 2050 strategy without an operator.
- **It is the only option with an anchor offtaker.** Rural energy projects in Japan typically fail their financials for lack of steady demand. The data center is the 24/7 baseload that makes the energy economics — and therefore the EV charging and blackout resilience — actually close. Solar-only and forestry-only have no anchor and stall.
- **It puts underused space to productive use.** Option A keeps paying upkeep on empty classrooms; Option E turns spare capacity into a revenue-generating facility (and, at the Koryukan, complements its cultural role).
- **It is the only option that does several jobs at once** — heals the forest, generates energy, creates higher-skill local jobs, and produces a replicable model — because the thinnings that fund the energy are the same thinnings that restore the forest.

**Honest counter-point:** Option E is also the **highest in cost, complexity, and execution risk.** The mitigant is structural, not rhetorical — the **funding-gate / descope mechanism** (§4) means the village never commits to the full build up front; each phase proceeds only if the prior gate's funding and feasibility clear. If the project stalls, it stops at a gate having delivered partial value, not at an idle building.

# 3. The ask, the offer, the proof — per audience

Each decision-maker needs the same underlying story ordered for *their* question: **Why you? → What I'm asked for → What I get → How I know you'll deliver.**

| Audience | What we ask | What they get | How they can be confident |
|---|---|---|---|
| **Village government** | Lease premises (former Sugano school / factory — one of several candidate sites); endorse as 官民連携 partner; co-apply for 交付金 | Depopulation reversal, 8–15 jobs, RE-plan delivered, underused space activated | Staged gates (no lock-in); open data; feasibility studies before any build |
| **MoE / grant bodies** | Award 交付金 / planning grants | Adopted RE plan actually implemented; carbon; a replicable 過疎 model | Plan already adopted (step 1 done); milestone-based disbursement |

| Audience | What we ask | What they get | How they can be confident |
|---|---|---|---|
| **Equity investors (Tier 2)** | Minority capital into the GK opco | Contracted share of operating surplus (net ¥10–32M/yr by Y5); green-compute exposure | Two-tier structure; anchor offtaker; payback model; descope safety |
| **Corporate partners** | Hosting offtake / CSR / equity | Cheap green compute, ESG story, or contracted return | Working pilot before scale; named advisory bench |
| **Foundations** | Mission-restricted grant | Measurable SDG/impact outcomes, strong narrative | Open methodology; transparent reporting |
| **Forestry co-op / landowners** | Sugi harvesting access; restoration participation | Fair harvest compensation; J-Credit income; reduced wildlife damage | Phase-1 contracts; village backing |

**Proof of delivery capability** (the "how do I know" answer, assembled): founder's Safecast track record; named advisory bench (Ray Ozzie); the village's *already-adopted* RE plan; the open-data / open-methodology commitment; and the gate structure that makes every commitment conditional and reversible.

# 4. Whole-project return, sensitivity & go/no-go

## 4.1 Scenario P&L by Year 5 (illustrative)

| | Low | Base | High |
|---|---|---|---|
| Annual revenue | ¥28M | ¥45M | ¥67M |
| Operating cost (~60%) | ¥18M | ¥27M | ¥35M |
| **Net surplus** | **¥10M** | **¥18M** | **¥32M** |
| Payback (¥200–290M capital) | ~18 yr | ~14 yr | ~10 yr |

## 4.2 The three swing variables (what would break the ROI)

The return is most sensitive to three figures, all resolved in Phase 1–2 feasibility:

1. **Data-center occupancy / hosting price** — the anchor. If hosting demand is weak, the whole stack thins. *Mitigant: secure a corporate offtake LOI before Gate 3 build.*
2. **Biomass CHP feedstock cost & sizing** — if thinning logistics cost more than modelled, energy margin erodes. *Mitigant: independent feasibility early; forestry programme supplies fuel at internal cost.*
3. **FIT/FIP tariff outcome** — sets *surplus*-energy value only. *Mitigant: revenue is not FIT-dependent — it is one of six streams. But note the plant is an **exporter** (~1.15 MW, sized to the forest); the co-located data center is only a small behind-the-meter baseload (~2% of output). Most of the CHP's own output goes to the village + grid export, so at >1,000 kW that energy revenue is likely **FIP**, not the 40 円 FIT (confirm). Grid interconnection capacity (Henry's 事前相談) is therefore a real dependency. This is the plant's electrons only, though — it doesn't cap total hosting revenue, since the larger showcase DC at the old school runs on grid/green power, not CHP output (compute is worth ~6–20× more per kWh than FIP either way). Figures provisional — see `../energy-forest/mitsue_fit_grid_check.md`. *Net of real GPU capex, that 6–20× gross figure only converts to net profit above a breakeven fill-rate (~45–70% depending on price tier/depreciation life) — see `mitsue_revenue_model.md` §1a. The swing variable is therefore GPU utilization (the HIGHRESO offtake pipeline), not the FIT/FIP outcome itself.*

### 4.2a GPU server net profit — 3-tier utilization scenario

(Added 2026-07-17.) Point 3 above nets real GPU capex against the gross ~6–20× compute-vs-FIP figure — the result is fill-rate-gated, not a free multiple (full breakeven math in `mitsue_revenue_model.md` §1a). Rather than one number, here is the same **Low / Base / High** framing as §4.1, applied to a single 8×H200 anchor server (~¥53M capex, ~10 kW draw, list pricing ¥2.53M/mo, 20% opex — see §1a for sources):

| | Low | Base | High |
|---|---|---|---|
| Utilization (fill-rate) | 50% | 65% | 85% |
| Depreciation life | 4 yr | 4 yr | 6 yr |
| Server revenue (at utilization) | ¥15.2M | ¥19.7M | ¥25.8M |
| Opex (~20%) | ¥3.0M | ¥3.9M | ¥5.2M |
| Depreciation | ¥13.25M | ¥13.25M | ¥8.83M |
| **Net profit per server/yr** | **−¥1.1M** | **+¥2.5M** | **+¥11.8M** |
| vs. FIP-equivalent (~¥3.1M /yr, same kWh, ~zero capex) | worse by ¥4.2M | worse by ¥0.5M | **~3.8× FIP** |

**Reading it:** below ~65% fill-rate at list pricing, a GPU server is *no better than* simply exporting the same power via FIP — sometimes worse, once its own capex is serviced. The project only clears a clear win at the **High** tier, which needs both strong utilization *and* the 6-year hyperscaler depreciation standard. **The swing variable is fill-rate, and that number is not yet known** — it depends on what the HIGHRESO offtake relationship can actually underwrite. Treat anchor-compute sizing as gated by that answer, not by available CHP output.

### 4.3 Go/no-go is already built in

The project's decision spine **is** a go/no-go system — it is just not labelled as one. Each gate is a whole-project kill/continue point:

| Gate | Funding test | If short |
|---|---|---|
| G1 (after Phase 0) | ¥3–8M secured | Hold & re-pitch |
| G2 (after Phase 1) | ¥30–50M + feasibility clears | Hold / descope |
| G3 (after Phase 2) | ¥120–290M + offtake LOI | Stage the build |
| G4 (after Phase 3) | Revenue online? | Partial → stay in pilot |

**Kill criteria** beyond funding: no JP co-founder verbal commitment by end of Phase 0 → G1 held; building structurally unfit (Phase 1 survey) → redesign or stop; feasibility shows negative base-case surplus → descope to forestry+solar only.

## 5. Residual risk & key dependencies

The [implementation plan risk register]() lists 9 risks with mitigations. What it omits — and investors ask for — is the **residual** (what remains after mitigation) and a consolidated **dependency map**.

### 5.1 Residual risk (after mitigation)

| Risk | After mitigation, what remains |
|---|---|

| Risk | After mitigation, what remains |
| --- | --- |
| Founder dependency (Rob) | Reduced by funded Rep.-Director role + Article 8a succession, but a single-person loss in Phase 0–1 still sets the project back months. **Unmitigated tail.** |
| JP co-founder not secured | Gate-1 hold protects spend, but failure to find one *ends* the project. **Hard dependency.** |
| Hosting demand weak | Six revenue streams cushion it, but the anchor thesis weakens; project survives smaller, ROI lengthens. |
| Feedstock economics | Internal-cost fuel helps, but persistent high logistics cost permanently compresses energy margin. |

### 5.2 Key dependencies (in order)

1. **JP co-founder** — gates everything (legal entity, local trust, bus-factor).
2. **Village endorsement** — unlocks 交付金 and the building.
3. **交付金 award** — closes most of the funding gap.
4. **Corporate offtake LOI** — validates the anchor before the expensive build.
5. **Fiber/grid connection** — physical feasibility.

### 5.3 Key personnel & bus-factor

Currently the team's load-bearing point is the founder. Mitigations in place: funded Representative-Director role (removes unpaid-time dependency), Article 8a continuity/succession (interim Rep. Director appointed within 30 days of a key founder's loss; credentials and documented processes maintained). **Remaining exposure:** the JP co-founder seat is unfilled, and no second operational person yet exists — closing both is the top Phase-0 priority. *(A short positive team-profile — who the principals are and why they are credible to investors — should be added once the JP co-founder is named.)*

## 6. Decision table — authority + investor evaluation

Two linked views: **who decides what (authority)**, and **what each decision-maker needs to see (evaluation)**.

| Decision | Authority | Gate / when | Evidence they need | What they get if yes |
| --- | --- | --- | --- | --- |
| Lease/endorse premises (former Sugano school / factory — site TBD Phase 1) | Village council / mayor | Phase 0–1 | Alternatives table; jobs forecast; site survey | Revitalization; RE-plan delivery; idle asset revived |
| Award 交付金 | MoE (via village) | Phase 1–2 | RE-plan fit; carbon; 過疎 eligibility | Adopted plan implemented |
| Commit as JP co-founder | The individual | Phase 0 (G1 trigger) | Charter; role; compensation (funded) | Founding equity + mission role |
| Invest equity in GK | Investor | Phase 2 (G3) | Two-tier structure; P&L; sensitivity; offtake LOI | Contracted minority return |
| Commit hosting offtake | Corporate partner | Phase 2–3 | Pilot performance; price/kWh; green cert | Cheap green compute / ESG |

| Decision | Authority | Gate / when | Evidence they need | What they get if yes |
|---|---|---|---|---|
| Supply feedstock | Forestry co-op / landowners | Phase 1 | Harvest terms; J-Credit share | Forest income; reduced crop damage |
| Proceed past each gate | Board (Tier 1) | G1–G4 | Funding secured + feasibility/kill-criteria | Authorisation to spend next phase |

---

## Sources & status

Financial ranges consolidate the project's own illustrative estimates from `mitsue_revenue_model.md`, `mitsue_implementation_plan.md` §ROI, and `mitsue_evm_plan.md` — not externally audited. 交付金 terms: MoE 地域脱炭素移行・再エネ推進交付金, 実施要領 (補助率 2/3・3/4) https://www.env.go.jp/content/900470616.pdf. Entity structure (一般社団法人 + 合同会社 GK) per `mitsue_implementation_plan.md` §1A and the NGO-setup checklists. **Items still to confirm:** final site selection from the candidate premises (former Sugano school spare classrooms and/or a disused factory — Phase 1; the dome school is a separate building, out of scope); equity entry valuation (Phase 2); positive team profile (post JP co-founder).

---

*Rob Oudendijk — YR-Design / Safecast · Mitsue, Nara Prefecture, Japan · June 2026*